HANDS-ON LAB REPORT

Hands-on Lab Report: Scanning Network Environment with Nmap GUI (Zenmap)

Skills Network Cloud Lab
12 October 2026

Executive summary

This lab used Zenmap, the graphical interface for Nmap, to perform Quick and Intense scans, examine port states and service fingerprints, and review topology and host details. Additional scans were performed against localhost and cloud.ibm.com. Results are recorded from the supplied screenshots.

Tooling: Nmap / Zenmap GUI (version displayed: 7.991), Npcap, Skills Network cloud-based Windows lab.

1. Objective

Use Zenmap, the graphical interface for Nmap, to scan hosts by domain name or IP address; review scan profiles; and interpret the port, service, host, and topology information returned by the tool.

2. Environment and tools

Platform: Skills Network cloud-based Windows lab environment.

Tool: Nmap with Zenmap GUI (version shown in the screenshots: Nmap 7.991).

Supporting component: Npcap, installed with the Windows package to support packet capture and related network functions.

Date performed: 12 October 2026.

3. Procedure

1. Opened the provided Windows lab environment and launched a browser.

2. Navigated to the official Nmap download site and selected the Windows installer.

3. Accepted the Nmap and Npcap license prompts and completed installation.

4. Launched Zenmap from the desktop.

5. Ran a Quick scan against scanme.nmap.org.

6. Ran an Intense scan against scanme.nmap.org to review additional service/version, OS-detection, and traceroute output.

7. Reviewed the Ports/Hosts, Topology, and Host Details tabs.

8. Ran a scan against 127.0.0.1 and a quick scan against cloud.ibm.com.

4. Results and observations

scanme.nmap.org — Quick scan: TCP/22 was reported open (SSH), TCP/80 open (HTTP), and TCP/25 filtered (SMTP).

scanme.nmap.org — Intense scan: TCP/22 was reported open with service fingerprint tcpwrapped; TCP/25 filtered; TCP/80 open with an Apache HTTP server version fingerprint; and TCP/9929 open (nping-echo). The scan also displayed an OS guess and traceroute results.

127.0.0.1 — The lab environment reported TCP/135 (MSRPC), TCP/445 (Microsoft-DS), and TCP/3389 (MS-WBT-Server) open. These are Windows-related services and should be exposed only when required and appropriately protected.

cloud.ibm.com — The quick scan reported TCP/80 (HTTP) and TCP/443 (HTTPS) open for the IP address resolved during the scan.

Zenmap's Ports/Hosts tab gave a concise port summary, the Topology tab displayed the route graph, and Host Details summarized host status and scan-derived information.

5. Interpretation

Open means a service accepted or responded to connection probing on that port. Filtered means Nmap could not determine the state conclusively because filtering or a lack of response interfered with the probe. A service/version label is a fingerprint, not proof of a vulnerability.

The same domain can resolve to different IP addresses over time or across network locations, and scan results can change. OS and service-version detection are estimates based on network responses and may be inaccurate.

The localhost results describe the provided virtual lab machine, not necessarily the user's physical computer.

6. Key learning outcomes

Installed and launched Nmap/Zenmap in a Windows lab.

Used Quick scan and Intense scan profiles and compared their output.

Identified open and filtered TCP ports and their likely services.

Explored service/version detection, OS guesses, traceroute, host details, and topology visualization.

Learned that scan output is contextual and should be interpreted cautiously.

7. Conclusion

The exercise demonstrated how Zenmap makes Nmap scanning more accessible through a graphical interface. It provided practical experience with target selection, scan profiles, port-state interpretation, service fingerprinting, and visualization of network paths.

8. Responsible use

Only scan systems you own or have explicit permission to assess. Public targets may publish specific scanning rules; follow those rules and the course instructions. This report records observations from the lab and does not claim that the scanned hosts are vulnerable.

Results summary

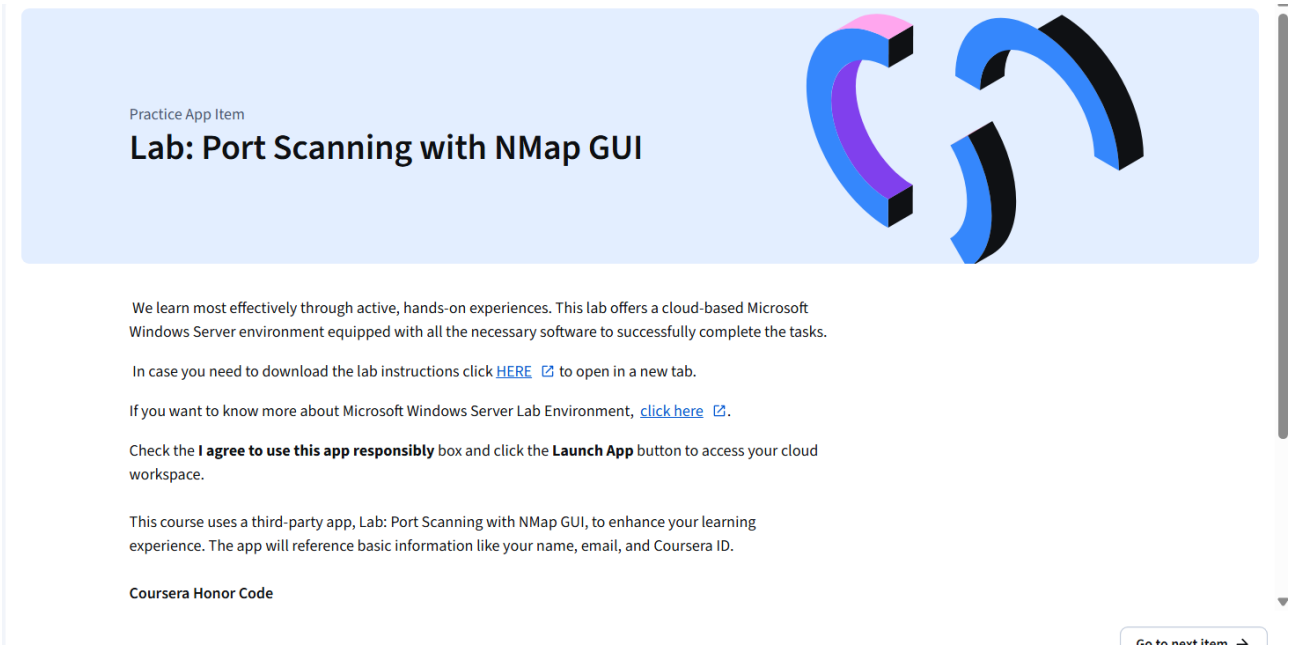
Target	Port	State	Service / note
scanme.nmap.org	22/tcp	Open	SSH; tcpwrapped label in intense scan
scanme.nmap.org	25/tcp	Filtered	SMTP
scanme.nmap.org	80/tcp	Open	HTTP; Apache fingerprint shown
scanme.nmap.org	9929/tcp	Open	nping-echo
127.0.0.1	135/tcp	Open	MSRPC
127.0.0.1	445/tcp	Open	Microsoft-DS
127.0.0.1	3389/tcp	Open	MS-WBT-Server
cloud.ibm.com	80/tcp	Open	HTTP
cloud.ibm.com	443/tcp	Open	HTTPS

Screenshot evidence

The following screenshots document the installation workflow and scan results.

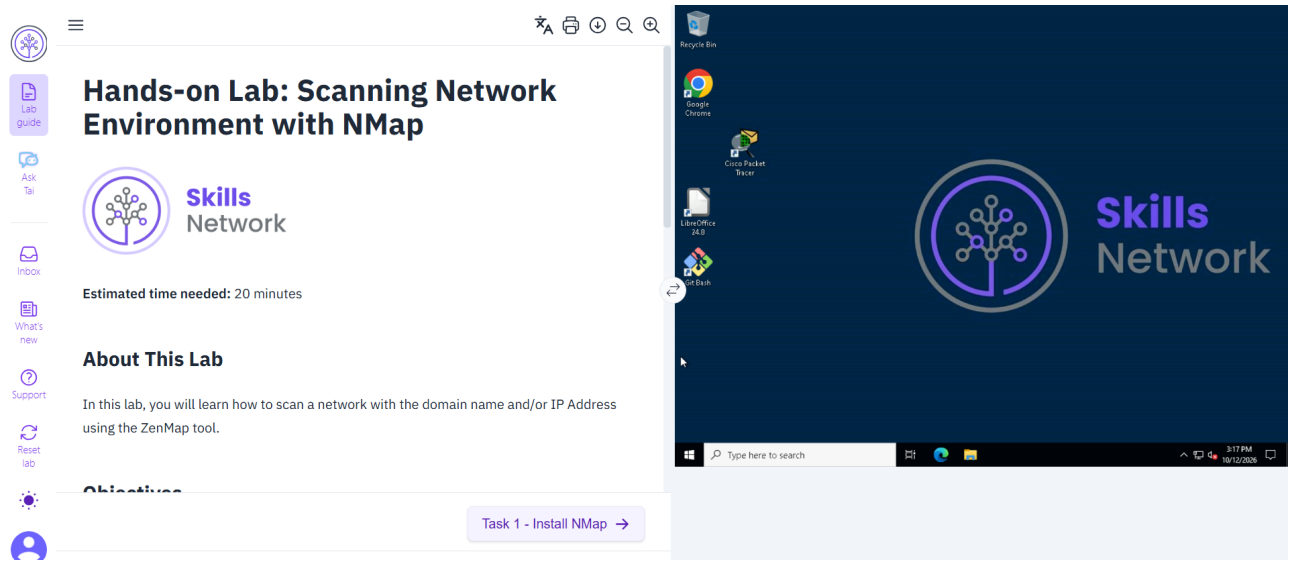
01. Lab landing page and objectives

The Skills Network lab introduces network scanning with Nmap and Zenmap.



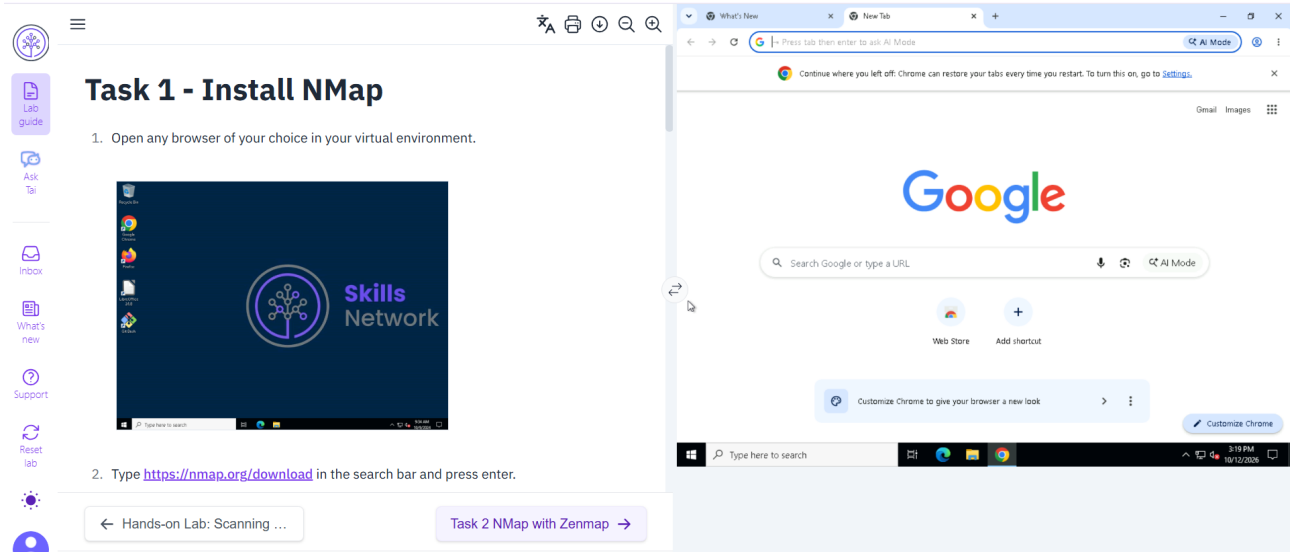
02. Cloud Windows lab environment

The Windows-based virtual environment is ready for the hands-on exercise.



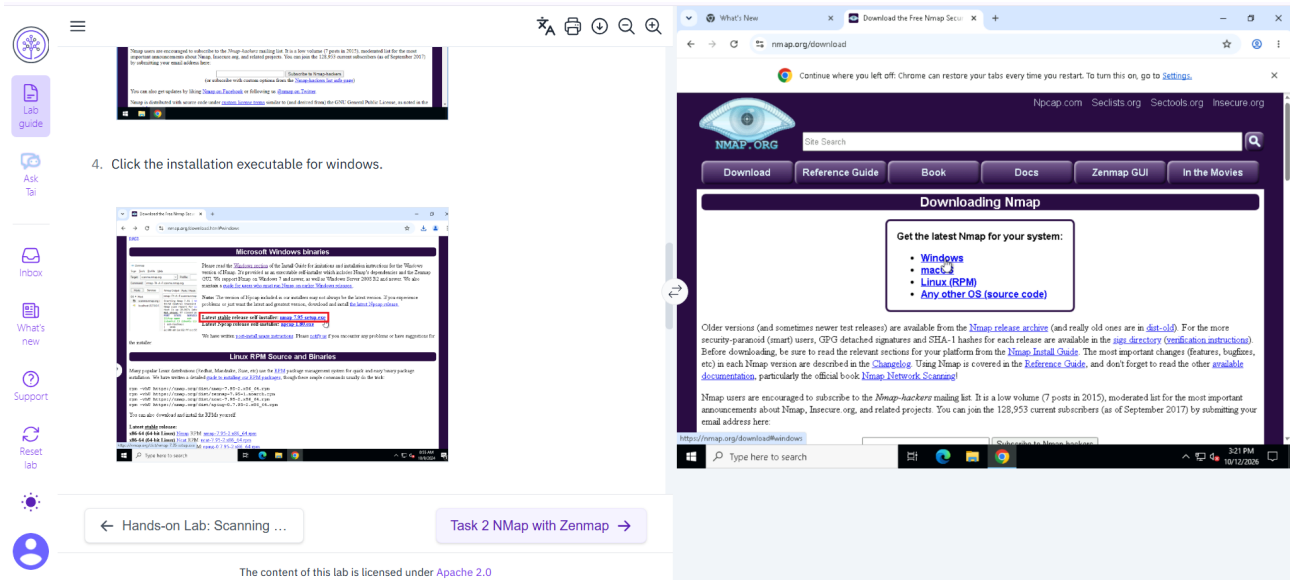
03. Open a browser

The instructions direct the learner to open a browser in the virtual environment.



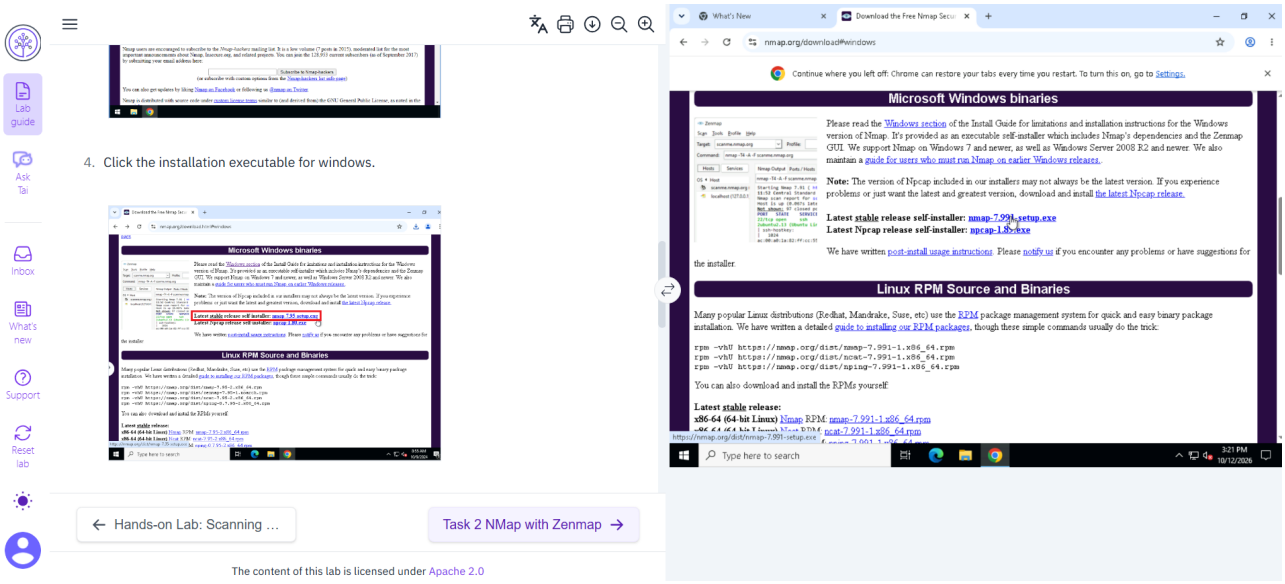
04. Nmap download page

The official Nmap download page is opened to obtain the Windows installer.



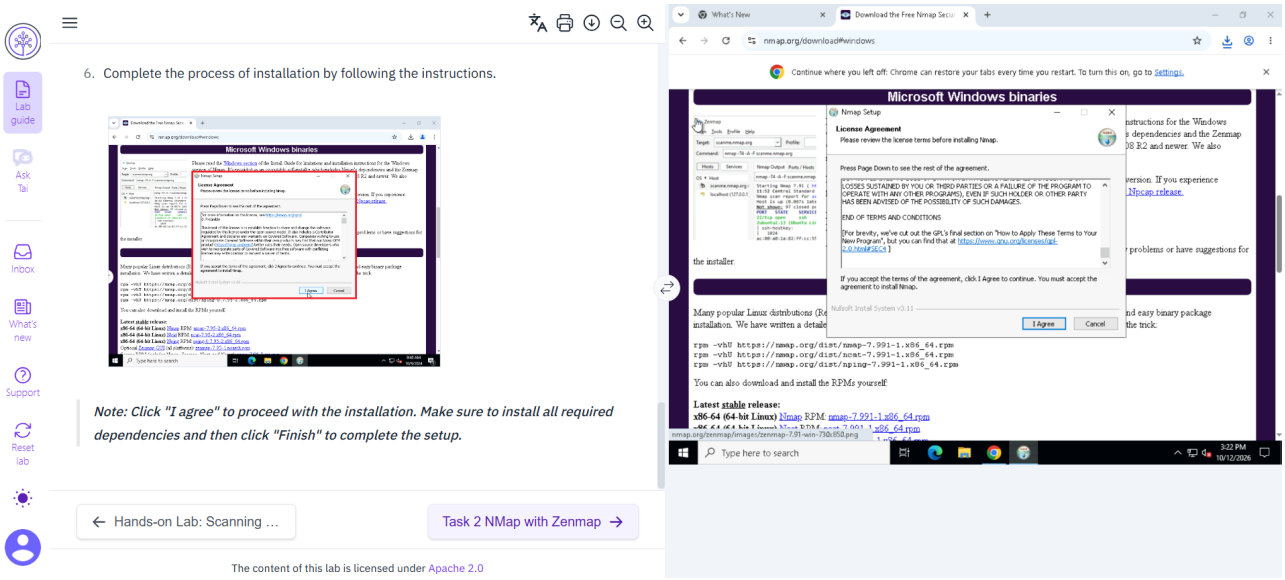
05. Windows installer selection

The Windows binaries page shows the Nmap self-installer and bundled Zenmap GUI.



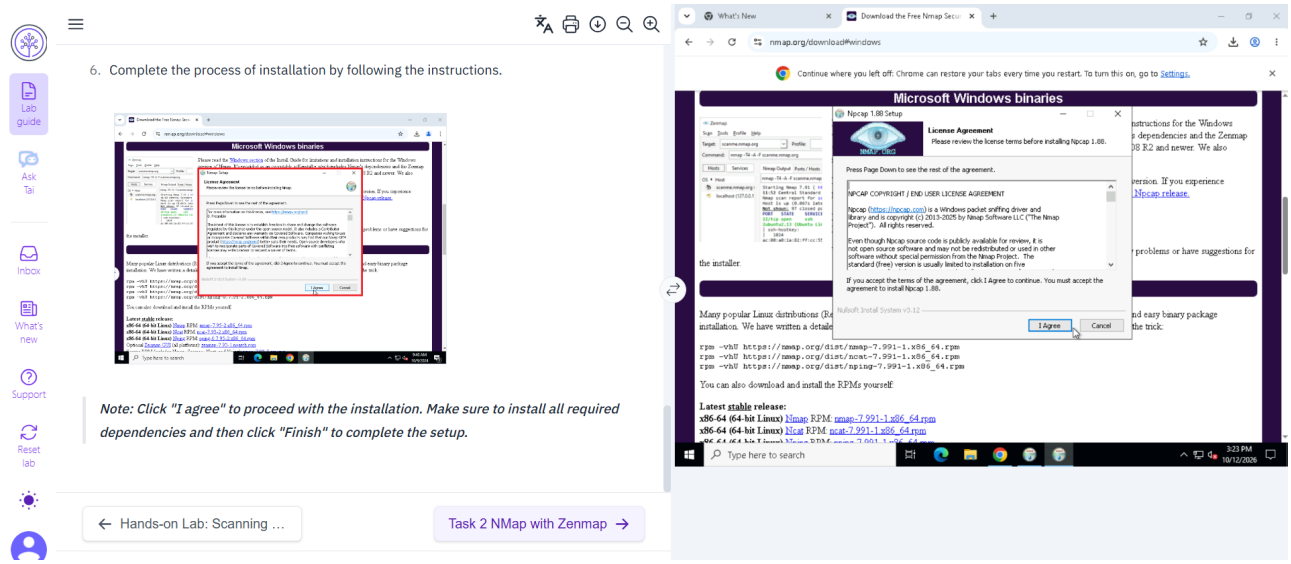
06. Nmap license agreement

The Nmap installer license agreement is displayed.



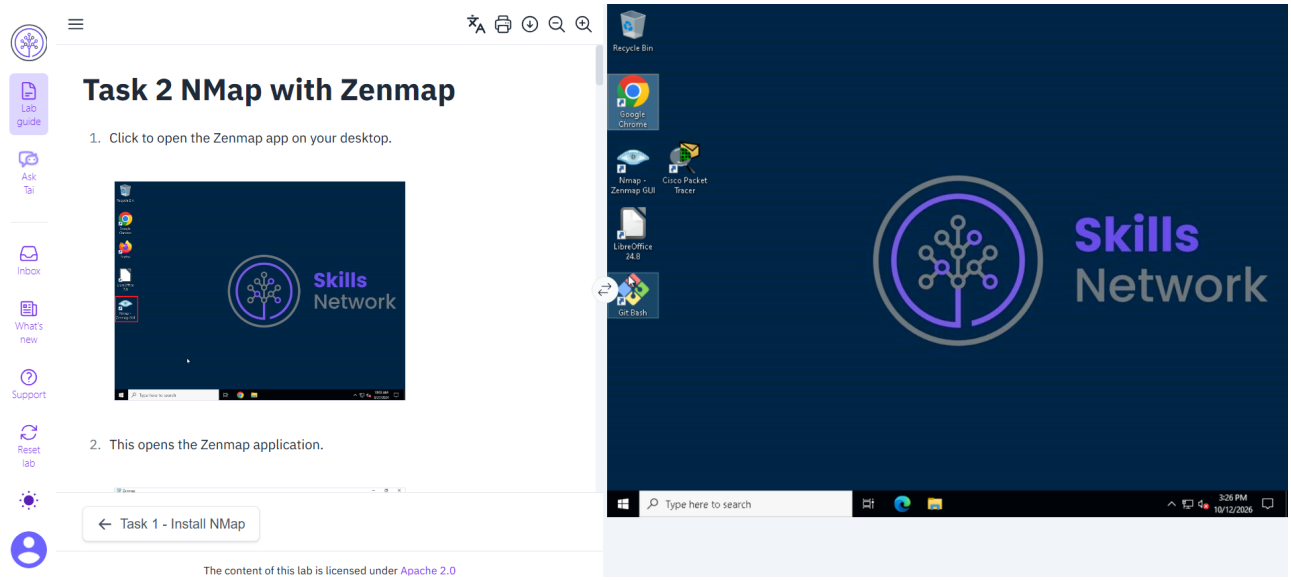
07. Npcap installation

The Npcap packet-capture driver license agreement is displayed as part of setup.



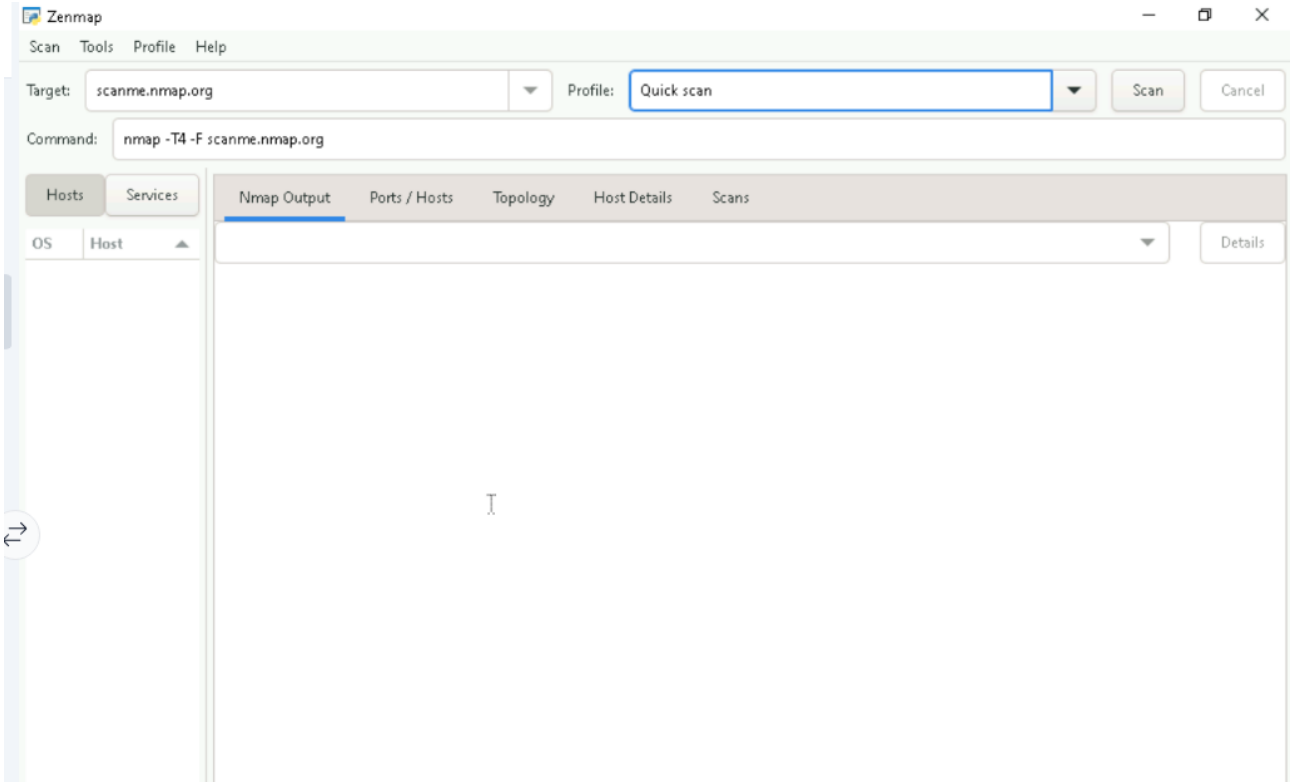
08. Launch Zenmap

Zenmap is available on the desktop after installation.



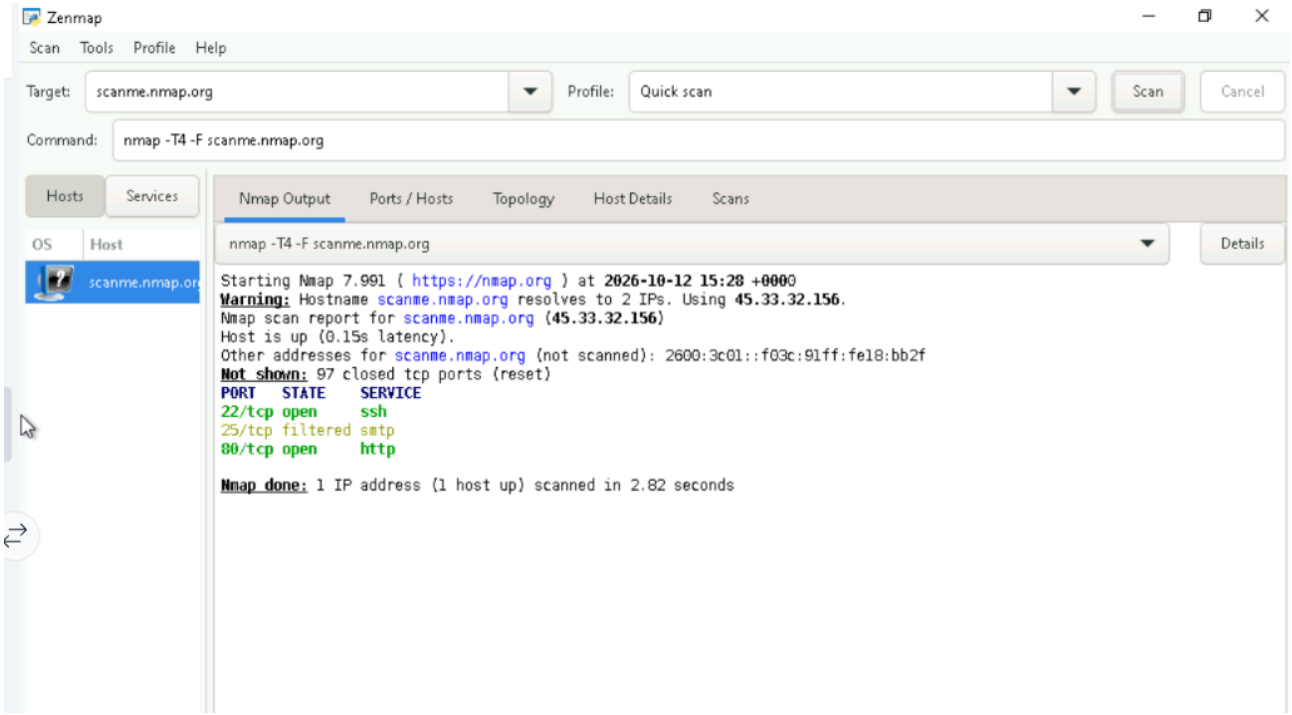
09. Zenmap quick scan configuration

The target is set to scanme.nmap.org and the Quick scan profile is selected.



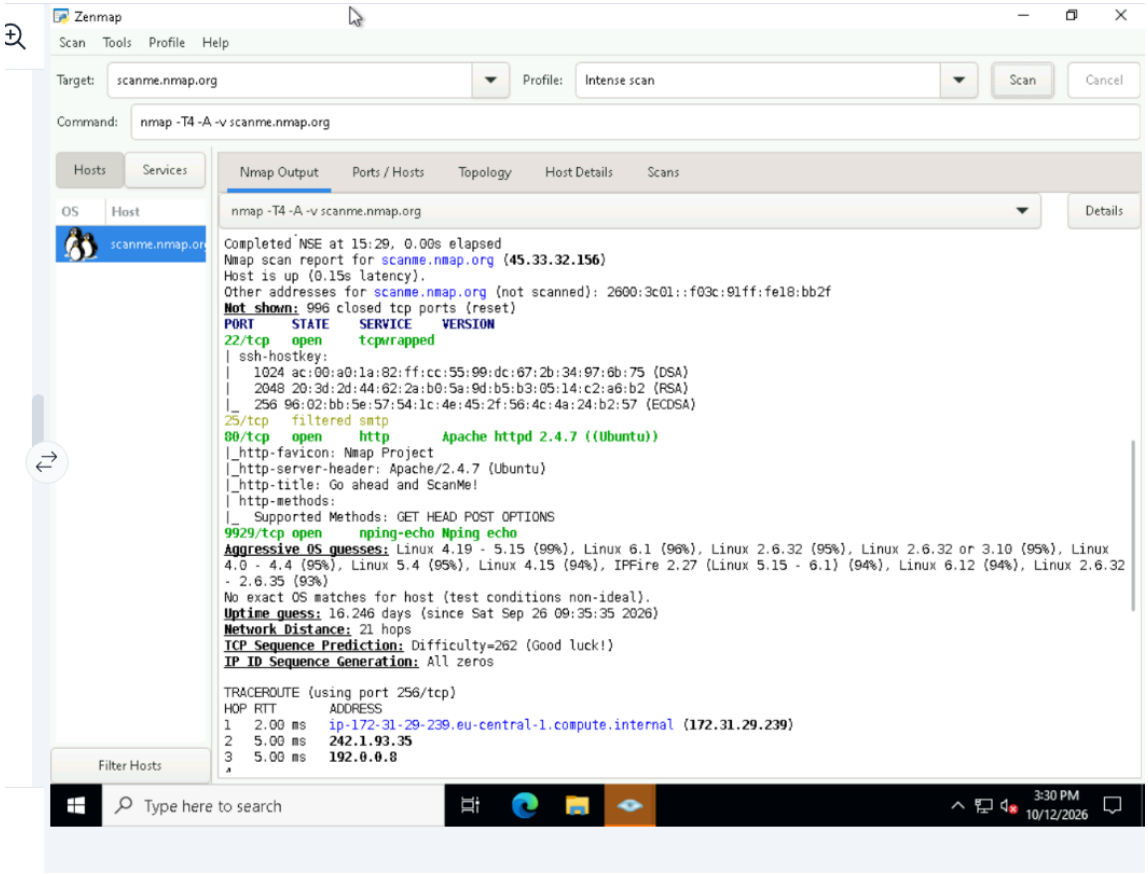
10. Quick scan results

The quick scan reports TCP/22 open (SSH), TCP/80 open (HTTP), and TCP/25 filtered (SMTP).



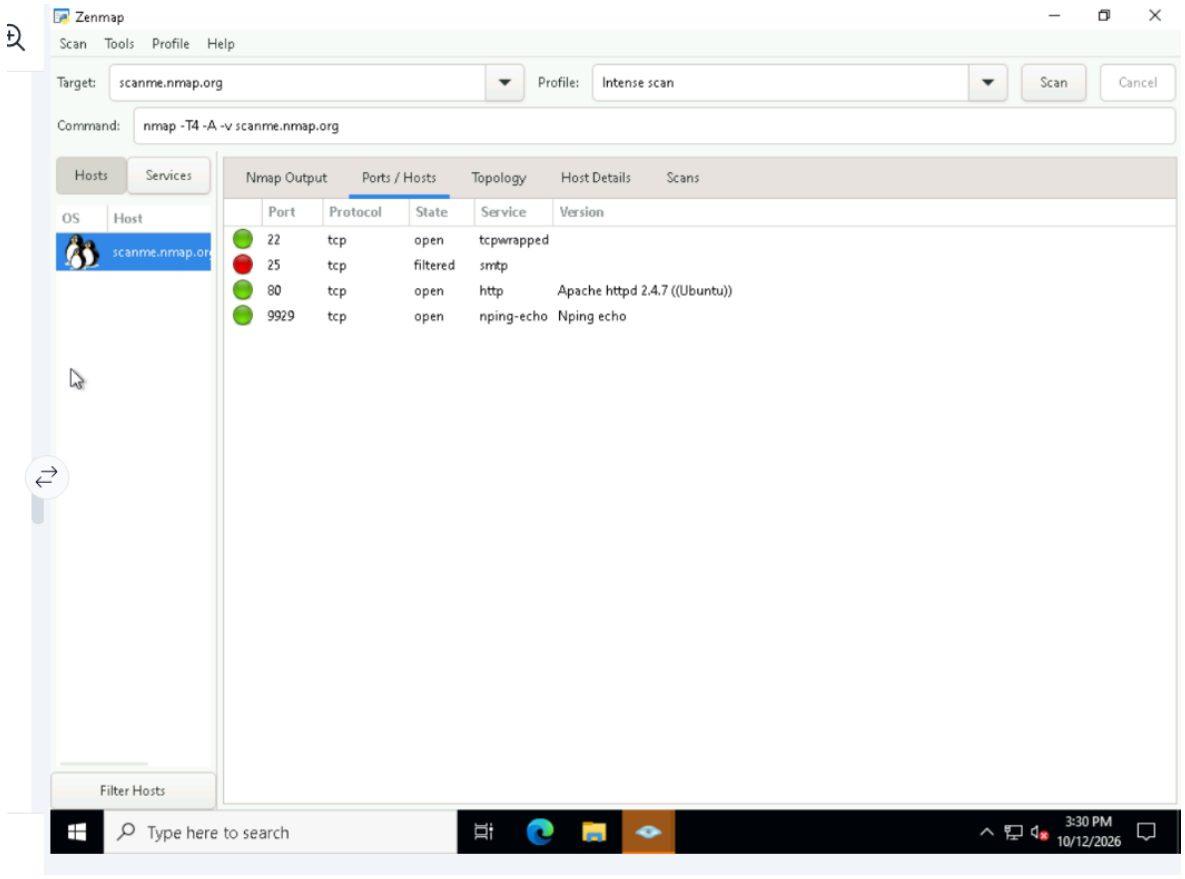
11. Intense scan results

The intense scan adds service/version detection and OS/traceroute information. It reports TCP/22 open (tcpwrapped), TCP/25 filtered, TCP/80 open (Apache HTTP server), and TCP/9929 open (nping-echo).



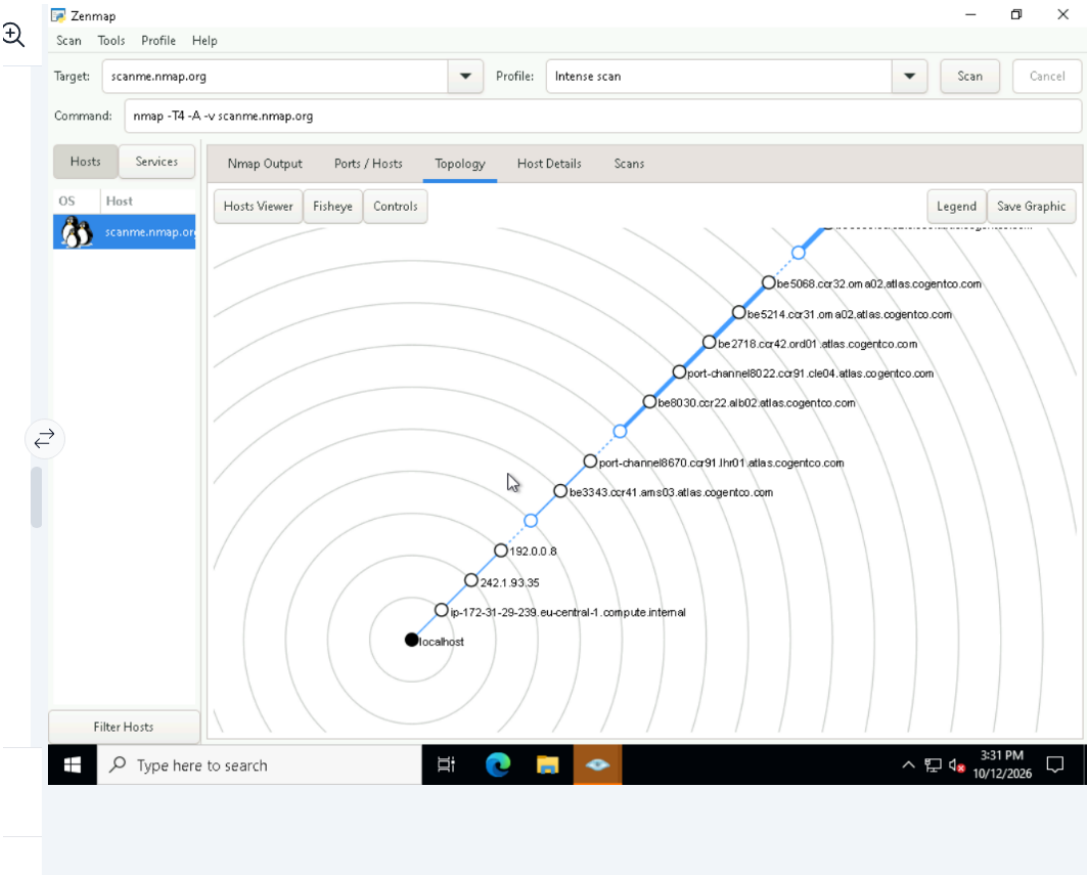
12. Ports and Hosts view

Zenmap's Ports/Hosts tab summarizes the discovered port states and services.



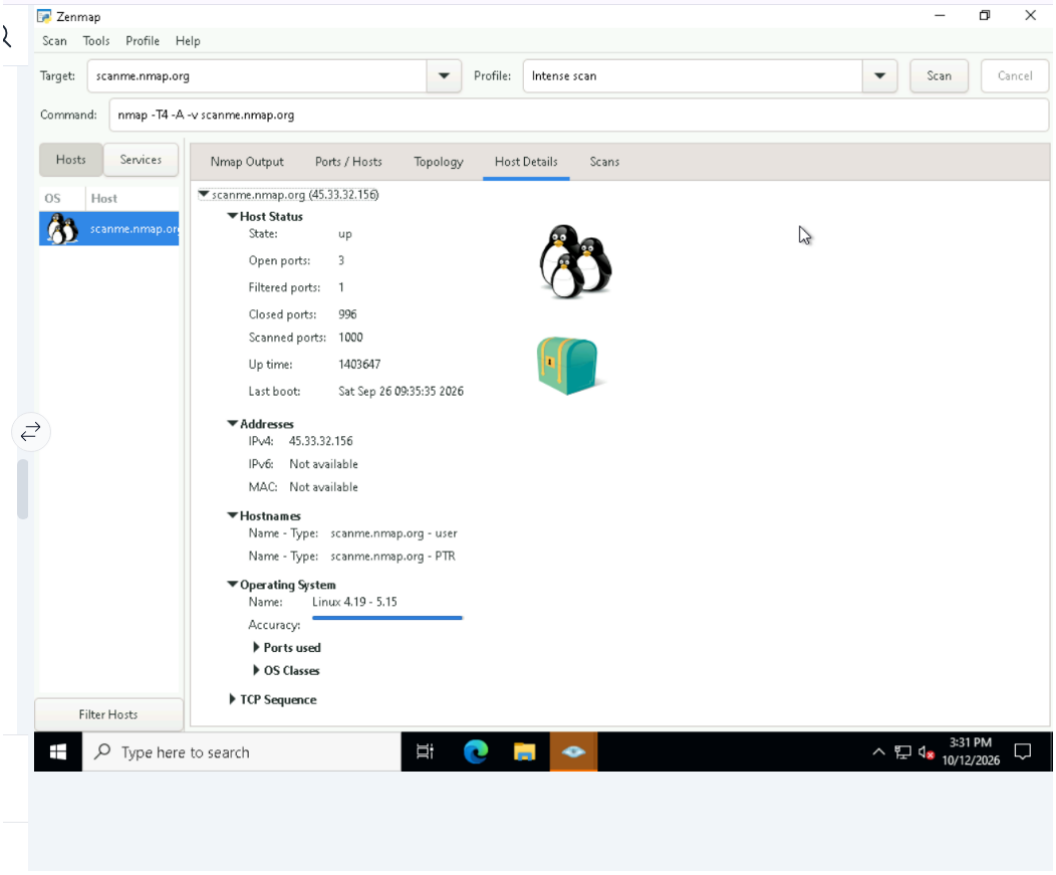
13. Network topology view

The topology view visualizes the route from the scanning environment toward the target.



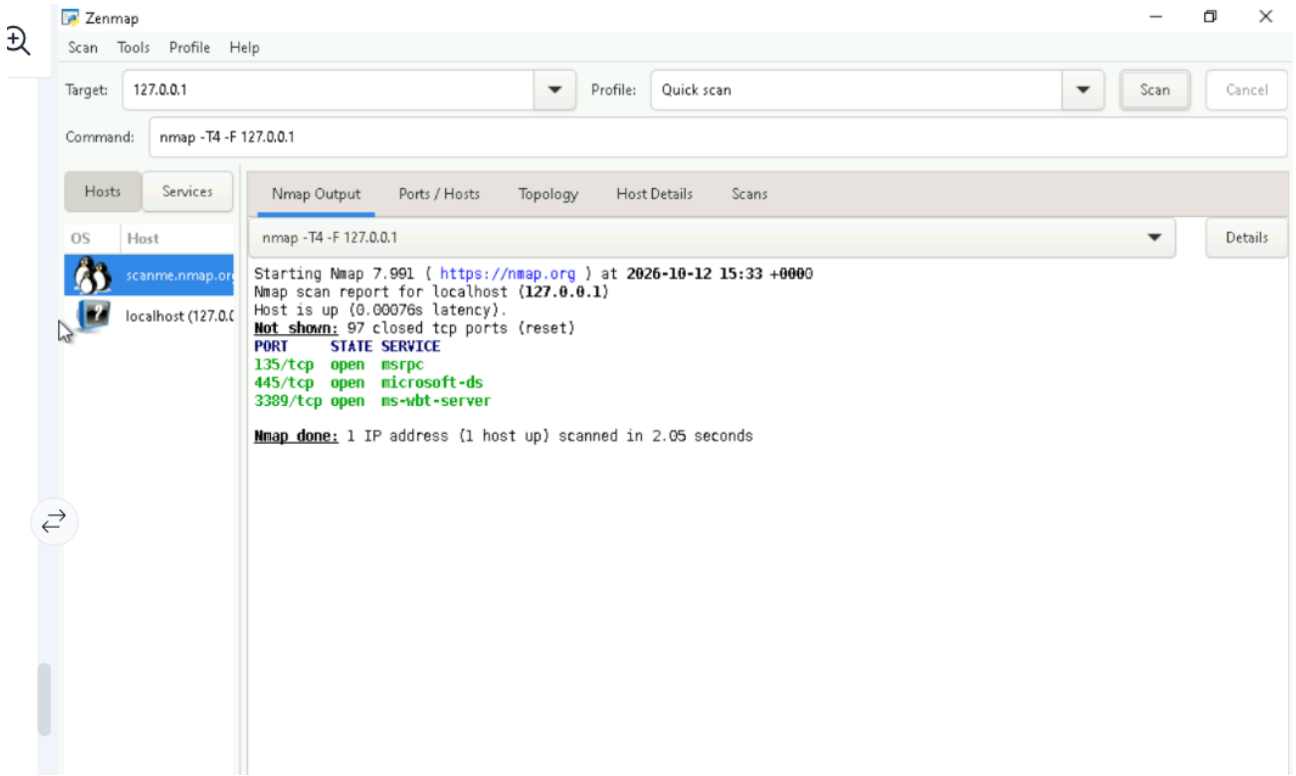
14. Host details view

Host Details summarizes host status, port counts, target address, and an OS guess.



15. Localhost scan

A scan of 127.0.0.1 reports TCP/135 (MSRPC), TCP/445 (Microsoft-DS), and TCP/3389 (MS-WBT-Server) open in this lab environment.



16. cloud.ibm.com scan

The quick scan reports TCP/80 (HTTP) and TCP/443 (HTTPS) open for the resolved cloud.ibm.com address.

